UNIVERSITY OF WEST LONDON

RESEARCH ETHICS AND INTEGRITY

RISK ASSESSMENT FORM

FORM 4

For all undergraduate and taught postgraduates Dissertations and Research Projects

FOR OFFICE USE ONLY

To be completed by module leader/supervisor *[subject to confirmation by SCREP]*

In the opinion of the module leader/supervisor this application falls into:

1a. Literature review – no risk	[ ]
1b. Literature review - some risk	[ ]
2a. Secondary data collection – no risk	[ ]
2b. Secondary data collection – some risk	[ ]
2c. Primary data collection – some risk	[ ]
2d. Primary data collection – high risk (send to UREC)	[ ]
3a Artefact - no risk	[✓]
3b. Artefacts - some risk	[ ]

Supervisor Name: Dr. Mohamed Shameem Signature: Date: 17/12/25

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Please fill in this form, then SAVE IT AS A PDF and submit as instructed by your supervisor with your project proposal

Your name: (first) **HESSA KAMAL** (last) **-**

Student number: **32146983** Your email address: **32146983@student.uwl.ac.uk**

FOR PROJECTS THAT INVOLVE MULTIPLE STUDENTS PLEASE ADD NAME, STUDENT NUMBER AND EMAIL ADDRESS OF EACH STUDENT

Name of supervisor: **Dr. Mohamed Shameem**

Title of project: ***Intelligent Cognitive Health Monitoring Through Ambient IoT and Immersive Visualization***

Date:**16 / 12 / 2025**

SECTION A

PROJECT DESCRIPTON

Please answer the following questions:

1. **Do you intend to involve human participants in the conduct of your research? If no, please skip questions 1a & 1b.**

   ☐ Yes ☐No✓

   1a. Does your research involve vulnerable adults (who are or may be for any reason unable to take care of themselves, or unable to protect themselves against significant harm or exploitation) or under-18s?

   ☐ Yes ☐No

   1b. Could your research potentially expose you, anyone assisting you, or participants to physical, psychological and/or emotional harm? (see Section B, Question 9)

   ☐ Yes ☐No

2. **Will your research involve travelling to geo-politically unstable regions/countries (e.g. areas affected by war, civil unrest, natural disasters, or listed as unadvisable to travel by the UK government)?**

   ☐ Yes ☐No✓

3. **Will your research involve access to security-sensitive material? (see the University's Research Ethics Code of Practice 2018 for a definition of security-sensitive materials and Section B, Question 9 of this form)**

   ☐ Yes ☐No✓

This proposal ***must be completed with the assistance of your supervisor*****/module leader/tutor**. You can change the size of the boxes (below) by typing or deleting as necessary.

It is very important to convey **with clarity**:

- Your research questions/the problem/the theme or topic you are investigating (what you are proposing to do and to find out or to create)
- The methodology or technical approach (for projects comprising in whole or in part the creation of an artefact) you will adopt – methods, number of participants, who the participants (if any) will be, survey instruments used, technology and equipment employed etc.; and what questions you are planning to ask your respondents (if applicable); how you will deal with technical challenges.

SECTION B

Only complete if you answered YES to Q1 in Section A.

	WHERE APPROPRIATE TO YOUR CHOSEN TOPIC/RESEARCH:		YES	NO	N/A
1	Will you describe in writing the main procedures to participants in advance, so that they are informed about what to expect? A copy of this must be attached to this application				✓
2	Will you tell participants that their participation is voluntary?				✓
3	Will you obtain written consent for participation and include within this that they have a right to withdraw at any point? A copy of this must be attached to this application				✓
4	If the research is observational, will you ask participants for their consent to being observed?				✓
5	With questionnaires, will you give participants the option of omitting questions they do not want to answer?				✓
6	Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs? This should be evidenced in the consent form and (if applicable) with a signed copy of UWL's data management form, attached to this application.				✓
7	Will you debrief participants at the end of their participation (i.e. give them a brief explanation of the study)? A copy of this must be attached to this application				✓
8	Will your project involve deliberately misleading participants in any way?				✓
9	If you answered YES to Question 1b (section A) give details on a separate sheet and state what you will tell your participants to do if they should experience any problems (e.g. who they can contact for help).				✓
10	Do participants fall into any of the following vulnerable groups? If they do, please and **tick box 2** overleaf. **Note that you may also need to obtain satisfactory DBS clearance (or equivalent for overseas students).**	Schoolchildren (under 18 years of age)			✓
		People with learning or communication difficulties			✓
		Patients			✓
		People in custody			✓
		People engaged in illegal activities (e.g. drug-taking)			✓
		Any other groups who could be reasonably argued as representing any form of vulnerability – please specify			✓

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SECTION C

	WHERE APPROPRIATE TO YOUR CHOSEN TOPIC/RESEARCH:	YES	NO	N/A
11	Will you be accessing materials which may be considered security-sensitive under the Counter Terrorism Act (2015)?		✓	
12	Does your project involve work with animals? If yes, please **tick box 2** below.		✓	

[Note: N/A = not applicable]

There is an obligation on the researcher to bring to the attention of the School Ethics Panel any issues with ethical implications not clearly covered by the above checklist.

PLEASE TICK **EITHER** BOX 1 OR BOX 2 BELOW AND **PROVIDE THE DETAILS REQUIRED** IN SUPPORT OF YOUR APPLICATION. THEN SIGN THE FORM.

Please tick

1.	I consider that this project has **no** significant ethical implications to be brought before the School Ethics Panel.	✓

2.	I consider that this project **may** have ethical implications that should be brought before the School Ethics Panel, and/or it will be carried out with children or other vulnerable populations.	

I have received guidance on ethical research practices relevant to my subject as part of my preparation for this module.

Signed ______________________ Print Name **HESSA KAMAL**

Date **16 / 12 / 2025**

(UG Researcher(s))

Signed Print Name **Dr. MOHAMED SHAMEEM**

Date **16 / 12 / 2025**

(Supervisor)

PROJECT OUTLINE

Your name: (first) **HESSA KAMAL** (last) **-**

Student number: **32146983** Your email address: **32146983@student.uwl.ac.uk**

Name of supervisor **Dr. Mohamed Shameem**

Title of project ***Intelligent Cognitive Health Monitoring Through Ambient IoT and Immersive Visualization***

Date: **16 / 12 / 2025**

Introduction to the research

Mental health challenges among students and professionals have reached crisis levels, with 77% of workers experiencing work-related stress. Traditional interventions require active engagement during moments when cognitive resources are depleted. Physiological computing offers continuous, passive monitoring through heart rate patterns that detect stress before awareness. Recent AI advances enable real-time pattern recognition on low-power devices, making privacy-preserving monitoring feasible. LSTM autoencoders demonstrate 85%+ accuracy in stress classification from HRV, though existing systems prioritize clinical diagnosis over everyday wellness and compromise privacy through cloud dependency. Ambient information visualization—presenting data through environmental changes rather than explicit displays—promotes behavioral change without cognitive overhead, aligning with "calm technology" principles ideal for stress management.

Research Questions:
1. Can multimodal IoT sensing detect cognitive stress before subject awareness?
2. Does ambient feedback improve stress management?
3. What visualization most effectively promote cognitive self-awareness?

Aims: Develop a functional IoT system integrating sensors, ML, and dual-mode ambient feedback; validate personalized LSTM models achieving ≥80% stress detection accuracy; evaluate ambient versus explicit feedback efficacy.

Hypotheses:
H1: Personalized LSTM autoencoders achieving ≥80% sensitivity and specificity versus population-based models.
H2: Ambient feedback will increase intervention engagement and reduce subjective stress versus explicit notifications.
H3: Multimodal sensing will provide early detection than heart rate alone.

Method

Research design or schedule (for creative artefacts)

Project Structure:
Dataset Selection & Preparation (Weeks 1-2)
Algorithm Development & Validation (Weeks 3-6)
Interface Prototyping (Weeks 7-10)
System Integration & Testing (Weeks 11-12)

Participants, including (where applicable) collaborators in the making of creative artefacts

N/A

Materials (to include locations and objects/resources)

Public datasets, RaspberryPi kit, relevant literature and books.

Procedure or details of technical aspects of creative production

N/A

Analysis

The project is an IoT-based cognitive wellness system that employs machine learning monitors stress through physiological and behavioral analytics like heart rate variability and keystroke dynamics, providing feedback via immersive 3D visualization and ambient lighting. This project develops and validates the system architecture using publicly available datasets rather than collecting new human participant data.

Feedback mechanisms include a morphing 3D "Mental Palace" environment that transitions from crystalline clarity to distorted architecture reflecting stress levels, and a "Cognitive Hue Lamp" that shifts colors along a blue-to-red spectrum. All processing occurs locally on edge devices (Raspberry Pi) ensuring privacy.

The project validates detection algorithms achieving ≥80% accuracy on existing datasets, develops functional prototype interfaces, and demonstrates proof-of-concept for continuous cognitive wellness monitoring. By combining AI, multimodal sensing, and ambient visualization, this addresses mental health challenges through self-awareness tools that promote early intervention without surveillance concerns.

INSTRUCTIONS TO SCREP/SCREP CHAIR

For applications 1a, 2a, 3a (no risk) – **Applications only need to be moderated by SCREP/SCREP Chair**

For applications 1b, 2b (some risks such as authorisations needed to access data or some emotional risk to students) - Supervisors commit to checking that authorisations have been granted and/or to keep in regular contact with student – **Applications only need to be moderated by SCREP/SCREP Chair**

For applications 2c, 2d and 3b (Some risks as involving human participants). A full application is necessary. **Applications need to be reviewed by SCREP panel and SCREP Chair to moderate**

Moderation is based on the sample

· being equal to 10% of submissions, or 10 submissions, whichever is the greater

· reflecting the whole mark range

· including all rejects and referrals